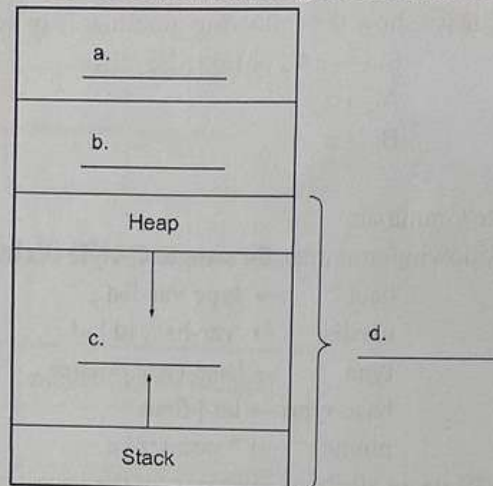


2024 NCKU CSIE Compiler Final Exam (total score: 100)

Student ID: _____ Name: _____

1. (10%) General Organization of Runtime Storage. Please fill in the blanks below.



2. (30%) For the following grammar:

- (a) $START \rightarrow EXP \$$
- (b) $EXP \rightarrow VAR$
- (c) $EXP \rightarrow EXP + VAR$
- (d) $VAR \rightarrow id$
- (e) $EXP \rightarrow (EXP)$

(You should follow “好康的pro max.pdf” methods and can not build or delete any rules.)

- a. (10%) Construct the DFA of LR(0).
- b. (10%) Construct the LALR(1) parsing table.

(State ID should corresponding to DFA in previous subquestion (a.))

The answer should be like below table:

	id	(	+	)	S	START	EXP	VAR
state 0								
state 1								
state...								

- c. (10%) Show the parsing stack and the actions of an LALR(1) parser for the input string “id + (id + id)”.

The answer should be like below table:

stack	input buffer	action (shift / reduce)

3. (20%) Ambiguous Grammar & Parser.

- a. (5%) Explain the definition of "ambiguity" for grammar.
- b. (5%) Please use a Venn diagram to illustrate the relationship among LR(0), LR(1), LALR(1) and SLR(1) parsers.
- c. (10%) Please show the following grammar is or is not LALR(1).

$$S \rightarrow aAd \mid bBd \mid aBe \mid bAe$$
$$A \rightarrow z$$
$$B \rightarrow z$$

4. (30%) Attribute Grammar.

Consider the following grammar for simple C-style declarations:

$$\text{decl} \rightarrow \text{type var-list ;}$$
$$\text{var-list} \rightarrow \text{var-list , id} \mid \text{id}$$
$$\text{type} \rightarrow \text{base-type pointer}$$
$$\text{base-type} \rightarrow \text{int} \mid \text{float}$$
$$\text{pointer} \rightarrow * \text{ pointer} \mid \epsilon$$

- a. (10%) Write an attribute grammar for the base-type of a variable.
- b. Draw the dependency graphs
 - i. (10%) corresponding to each grammar rule for the attribute grammar.
 - ii. (10%) for the declaration: **float * com , pile , r ;**

5. (10%) Give the sequence of three-address code instructions corresponding to each of the following arithmetic expressions:

a. $1+2+3+4$

b. $((1+(2+3)+4)+5$